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Question: Oral combinations of midazolam and ketamine (MIKE) compared to oral midazolam alone (MDZ) for preanesthetic medication in pediatric surgical patients

Setting: Pediatric Surgery

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Oral combinations of midazolam and ketamine (MIKE)	oral midazolam alone (MDZ)	Relative (95% CI)	Absolute (95% CI)		
Satisfactory sedation												
18	randomised trials	serious [a]	not serious	not serious	not serious	publication bias strongly suspected [b]	665/808 (82.3%)	507/748 (67.8%)	**RR 1.20** (1.10 to 1.31)	**136 more per 1,000** (from 68 more to 210 more)	⨁⨁◯◯ LOW	CRITICAL
Satisfatory Separation score												
16	randomised trials	serious [a]	not serious	not serious	not serious	publication bias strongly suspected [b]	541/649 (83.4%)	453/653 (69.4%)	**RR 1.20** (1.06 to 1.36)	**139 more per 1,000** (from 42 more to 250 more)	⨁⨁◯◯ LOW	CRITICAL
Satisfatory Mask Acceptance												
11	randomised trials	serious [a]	not serious	not serious	not serious	publication bias strongly suspected [c]	348/412 (84.5%)	305/420 (72.6%)	**RR 1.13** (1.04 to 1.24)	**94 more per 1,000** (from 29 more to 174 more)	⨁⨁◯◯ LOW	CRITICAL
Behavior at venepuncture												
9	randomised trials	serious [a]	not serious	not serious	not serious	publication bias strongly suspected [d]	292/394 (74.1%)	173/341 (50.7%)	**RR 1.32** (1.11 to 1.57)	**162 more per 1,000** (from 56 more to 289 more)	⨁⨁◯◯ LOW	CRITICAL
Anxiolysis												
5	randomised trials	serious [a]	not serious	not serious	not serious	none	192/248 (77.4%)	110/188 (58.5%)	**RR 1.20** (0.94 to 1.52)	**117 more per 1,000** (from 35 fewer to 304 more)	⨁⨁⨁◯ MODERATE	IMPORTANT

CI: Confidence interval; **RR:** Risk ratio

Explanations

a. Most studies classified as having high risk or multiple "some concerns" in one or more domains of the Cochrane Collaboration RoB 2 tool. b. Funnel plot asymmetry, Harbord's test $p < 0.001$ and multiple missing studies filled by Duval and Tweedie's trim-and-fill method.

c. Funnel plot asymmetry, Harbord's test $p = 0.07$, but multiple missing studies filled by the Duval and Tweedie's trim-and-fill method. d. Funnel plot asymmetry, Harbord's test $p = 0.04$, and multiple missing studies filled by the Duval and Tweedie's trim-and-fill method.